



Smart Local Bus Route Finder

A Database-Driven Analytical Dashboard for Karachi's Public Bus Network

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Database Systems Final Project · SZABIST University, Karachi

<https://smart-bus-route-finder.vercel.app>

Why this project

- Millions ride Karachi's buses daily, but there is no single system telling them which bus to take.
- Passengers do not know transfer points, fares, or travel time in advance.
- Operators have no consolidated view of ridership, revenue, reliability, or feedback.
- Route information is scattered across informal sources instead of one database.

A three-tier web application

- **Database:** PostgreSQL on Neon, 21 related tables
- **Backend:** Node.js + Express REST API (JSON)
- **Frontend:** HTML, CSS, Bootstrap, JavaScript
- **Charts:** Chart.js on a single dashboard page
- **Map:** Leaflet + OpenStreetMap route view

21

database tables

20+

REST API endpoints

13

analytical charts

Built with

PostgreSQL 16

Neon (cloud DB)

Node.js

Express.js

node-postgres

HTML5

CSS3

Bootstrap 5

JavaScript (Fetch / AJAX)

Chart.js

Leaflet

Vercel

Git / GitHub

Data coverage

- BRT: Green Line, Red Line, Orange Line
- Feeder: Peoples Bus Service R-10, R-12
- Electric: EV-1, EV-3 · shared interchange stops enable transfers

21 tables, fully constrained

- **Network:** zone, operator, route_type, route, stop, route_stop, fare_stage
- **Fleet and ops:** bus_model, bus, driver, staff, schedule, trip_log, maintenance
- **Passengers and revenue:** passenger, fare_card, card_transaction, journey, route_search, feedback, incident

22,000+

sample rows seeded

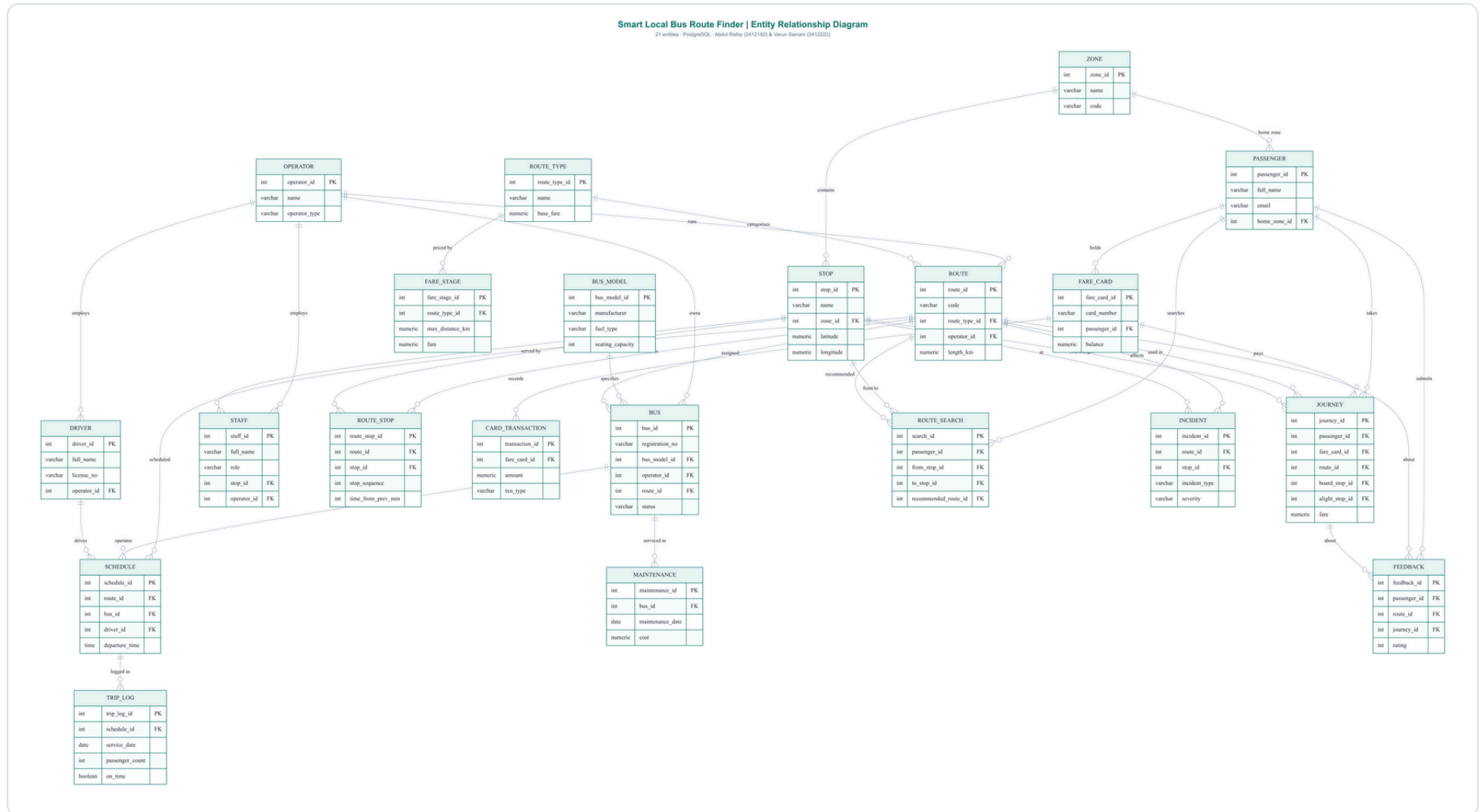
6,000

passenger journeys

PK / FK

+ check constraints throughout

ENTITY RELATIONSHIP DIAGRAM



Endpoints return JSON

Group	Examples
Analytics (13)	/api/analytics/summary, ridership-by-route, peak-hours, on-time-performance
Route finder	/api/finder/stops, /api/finder/plan?from=&to=
CRUD (8 resources)	GET, POST, PUT, DELETE on routes, stops, passengers, ...

- Connects to PostgreSQL through a pooled connection
- Performs CRUD and fetches analytical / statistical data

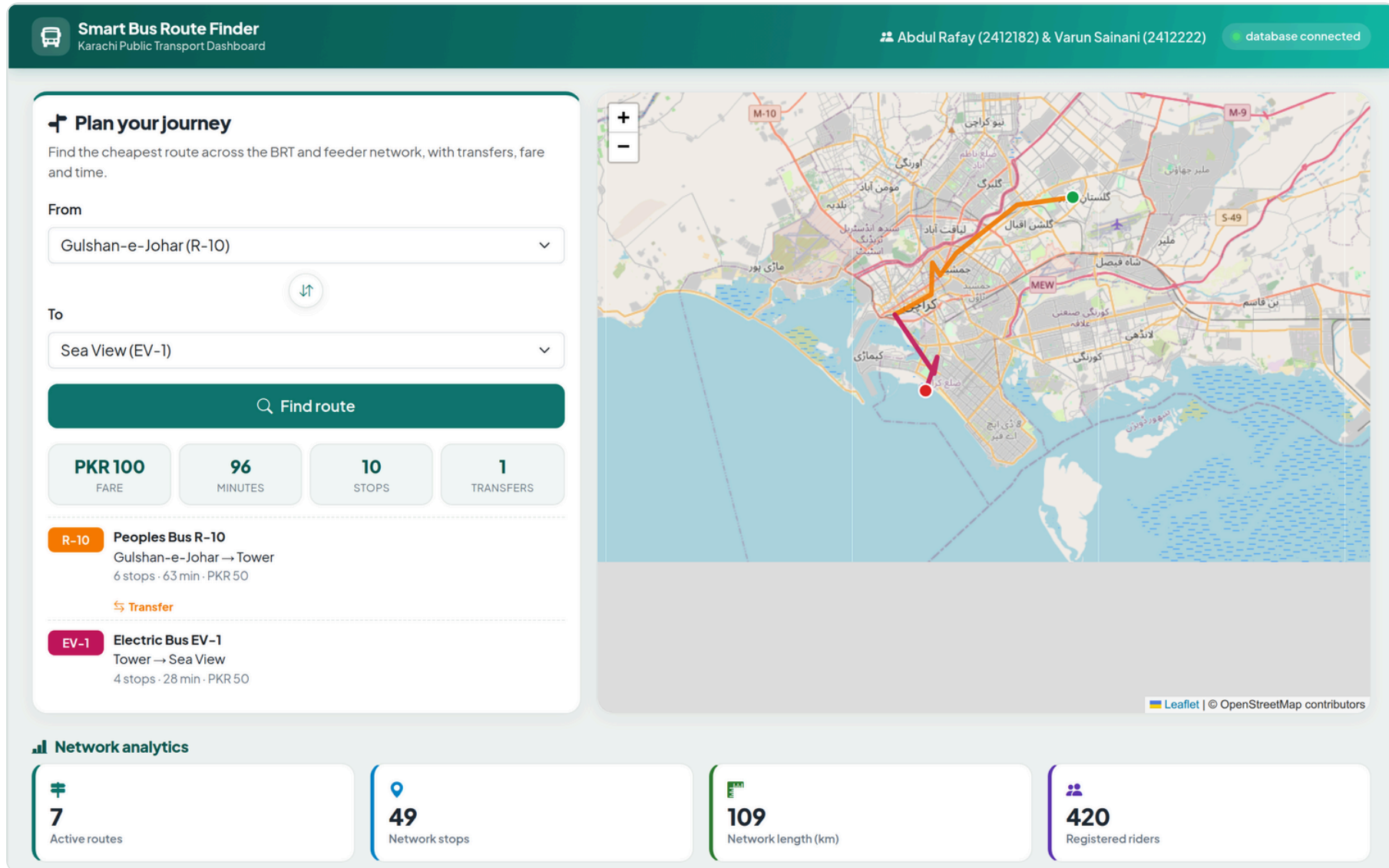
Dijkstra shortest path

- The network is a graph of (route, stop) states.
- Riding moves along a route; a transfer switches routes at a shared stop with a time penalty.
- Dijkstra's algorithm finds the lowest total travel time.
- Fare for each leg is computed from the fare-stage table; results show legs, fare, time, stops, and transfers.

Gulshan-e-Johar → Sea View

R-10 to Tower, transfer to EV-1: PKR 100, 96 min, 1 transfer

ANALYTICAL DASHBOARD



Network analytics

7
Active routes

49
Network stops

109
Network length (km)

420
Registered riders

ROUTE FINDER AND MAP

Plan your journey

Find the cheapest route across the BRT and feeder network, with transfers, fare and time.

From

Gulshan-e-Johar (R-10)

To

Sea View (EV-1)

Find route

PKR 100
FARE

96
MINUTES

10
STOPS

1
TRANSFERS

R-10

Peoples Bus R-10
 Gulshan-e-Johar → Tower
 6 stops · 63 min · PKR 50

Transfer

EV-1

Electric Bus EV-1
 Tower → Sea View
 4 stops · 28 min · PKR 50

Leaflet | © OpenStreetMap contributors

TESTING

Verified end to end

Test	Result
Health and database connectivity	Passed (200)
13 analytical endpoints	Passed (valid JSON)
CRUD round trip (POST, GET, PUT, DELETE)	Passed (201, 200, 200, 200)
Route finder with transfer	Passed (correct fare and time)
Invalid input handling	Passed (400 / 404)

Thank you

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